Portfolio Performance and Risk Analytics

Independent project · Python, pandas, NumPy, SQLite, Excel · eight USD ETFs, daily data 2021-2025 · retrospective simulation

Question

A diversified eight-ETF 60/40 portfolio returned 34.81% while a 60% VTI / 40% BND benchmark returned 44.68%. Where did the 9.87 pp gap come from, what risk did the portfolio take, and what do factor models say about it?

What the project does

A reproducible return engine with drifting weights, monthly rebalancing and 0/5/10 bps cost scenarios; holdings-based Brinson-Fachler attribution with Carino linking; Euler risk and tracking-error decomposition; 1-day VaR/ES (historical, normal, EWMA) backtested with Kupiec and Christoffersen tests; Fama-French five-factor plus momentum regressions with Newey-West errors; and a checksum-verified data contract with a separate data-quality pipeline.

Results

	Cumulative	Annualised	Volatility	Max drawdown	Sharpe	Tracking error
Main	34.81%	6.18%	10.63%	-22.22%	0.62	2.27%
Benchmark	44.68%	7.70%	11.18%	-21.49%	0.72	0.00%

• **Attribution.** Allocation 0.00 pp, selection -9.87 pp: both portfolios start each month at 60/40, so the gap is inside the sleeves. Equity sleeve -9.49 pp versus VTI (VB -3.41 pp, VEA -3.53 pp, VWO -2.55 pp).

• **Risk.** Equities produce 91% of volatility on a 60% weight; the VTI underweight drives 86% of tracking error. At 99%, VaR models under-cover over 1005 out-of-sample days (EWMA 17, historical 20, normal 23 exceptions vs 10 expected).

• **Factors.** US market beta 0.55 vs 0.61. Of the -1.48% a year average active return, market beta -0.58%, size -0.27%, profitability -0.16%, value +0.22%; residual alpha -0.67% (t -0.9) is not significant.

How the numbers are checked

26 unit tests with hand-computed expectations and look-ahead perturbations; SQLite recomputes all asset returns; an independent units ledger recomputes NAV for 12 scenarios; native-formula Excel workbooks reproduce the attribution; OLS matches numpy to 1e-15; identical results on Windows and Linux; prices from the data pipeline reproduce all 117 metrics within 3.1e-08.

Limitations

One five-year window chosen in hindsight; two-segment attribution because the benchmark has no international or small-cap sleeve; realised rather than forecast risk; US factors for international and bond ETFs; adjusted prices as a total-return proxy; no taxes or slippage.

Data: Yahoo Finance adjusted closes; Kenneth R. French Data Library factors. Data used for personal study and not redistributed. Not investment advice.